

Goal: To validate the Adaptive Integrity Model (AIM)—which is a model designed to explain, predict, and detect p-hacking in academic research. You'll need to use the following 5 publicly accessible datasets to validate AIM and the overall effectiveness of each of its 3 components:

- [\[1\] "Just Posting It" works, leads to new retraction in Psychology](#)
- [\[109\] Data Falsificada \(Part 1\): "Clusterfake"](#)
- [\[110\] Data Falsificada \(Part 2\): "My Class Year Is Harvard"](#)
- [\[111\] Data Falsificada \(Part 3\): "The Cheaters Are Out of Order"](#)
- [\[112\] Data Falsificada \(Part 4\): "Forgetting The Words"](#)

Note#1: What's great about the website where I found these datasets (in the “fake data section” is that the statisticians who have detected frauds in empirical research provide links to the original datasets, the published studies, and OSF pages where all the files used during pre-registration of retracted studies are available.

The Three Components of AIM:

1. Explanatory Component (Ip)

- Purpose: Quantifies the likelihood of p-hacking by accounting for external pressures and intrinsic biases.
- Formula: $I_p = 0.5 \times P_e + 0.5 \times B_i$

Variables

- P_e (External Pressures): Represents external factors such as the pressure to publish in high-impact journals or the need to secure funding.
- B_i (Intrinsic Biases): Reflects personal or career incentives, such as confirmation bias or the pressure to advance academically.

Constants

- **0.5:** This weight assigns equal importance to external pressures (P_e) and intrinsic biases (B_i) in contributing to p-hacking likelihood. These weights can be adjusted in future refinements.

2. Predictive Component (Ph)

- Purpose: Predicts p-hacking by identifying statistical anomalies, such as clustering of p-values near 0.05 or discrepancies between tests conducted and reported.
- Formula:
$$P_h = 1 \text{ divided by } 1 + e^{-(1 \times (T_c - T_r) + 2 \times \delta_p)}$$

Variables:

- T_c (Total Tests Conducted): The number of statistical tests performed during the research.
- T_r (Total Tests Reported): The number of tests included in the final publication.
- δ_p (P-Value Clustering): The proportion of p-values in the range 0.04–0.05, signaling potential p-hacking.

Constants:

- 1: Balances the weight of test discrepancies ($T_c - T_r$) in the formula.
- 2 : Assigns more weight to p-value clustering (δ_p), as it is a stronger signal of p-hacking.
- $1+e$: Normalizes the output to ensure the probability (P_h) is between 0 and 1.

3. Detection Component (T_x)

- Purpose: Assesses transparency and replicability in research.
- Formula: $T_x = (1 \times R_a + 2 \times V_r + 1 \times M_c)$ divided by 4

Variables

- R_a (Ratio of Reported to Conducted Analyses): Measures the proportion of tests disclosed compared to those conducted.
- V_r (Validation Robustness): Indicates replication success or robustness of the study when independently validated.
- M_c (Pre-Registration Match): Assesses alignment between pre-registered protocols and reported results.

Constants:

- 1 and 2: Weights emphasize the importance of each metric, with V_r weighted higher (2) due to its stronger signal of research reliability.
- 4: Normalizes the combined score to ensure consistency across datasets.

4. Combined AIM Integrity Score ($P_{integrity}$)

- Purpose: Integrates the three components into an overall research integrity score.
- Formula: $P_{integrity} = 1$ divided by $1+e^{-(I_p + P_h + T_x)}$

Explanation:

- Combines the outputs of I_p , P_h and T_x into a single score.
- $1+e$ ensures the score is normalized between 0 and 1, making it interpretable as a probability of research integrity.

5. Project Deliverables

1. Validation report need to includes:
 - Dataset summaries.
 - Calculations for I_p , P_h , and T_x and $P_{integrity}$.
 - Analysis of AIM's performance using sensitivity, specificity, and F1 Score.
2. **Code and Analysis Scripts:**
 - Provide R scripts used for calculations.
3. **Summary model validation tables**
 - Feel free to use as many tables as possible. But be sure to use APA formatting for every summary model validation table.

To give you a deeper understanding of and more context about this project, I've prepared following research summary. Please be sure to read it.

When researchers seek to publish in a leading peer-reviewed journal and the p-value of their study falls short of significance thresholds, they face an ethical dilemma: be honest and risk rejection or manipulate the p-value to secure publication. This study introduces the Adaptive Integrity Model, a new framework designed to explain, predict, and detect p-hacking. Unlike existing models such as P-Curve Analysis and Z-Curve, which focus essentially on detecting statistical anomalies, the Adaptive Integrity Model encompasses an explanatory component, which quantifies the external pressures and intrinsic biases that lead to unethical decisions; a predictive component, which identifies such statistical red flags as p-value clustering and test discrepancies; and a detection component, which assesses data transparency through reported analyses, replication success, and pre-registration compliance. Given its combinatorial design, The model resolves the limitations of P-Curve Analysis and Z-Curve and produces a research integrity score, which could fundamentally reshape the assessment and detection of p-hacking.

Note: The validation report does not have a minimum or maximum number of pages, but you're encouraged to include one or more APA-formatted summary result tables in it. As well, you are allowed to define variables such as Bi (intrinsic bias), which might reflect personal motivations that could influence p-hacking. In this case, you could, for example estimate its values by using mean emotional response variables like shame, guilt, embarrassment, fear, regret, anxiety, and anger, and divide by 10 to normalize the score between 0-1 (assuming score are rated on a 0-10 scale).